Skip to content

- Post category:Class 7 Notes

Class 7 Math Notes In English Chapter – 01 Integers available here in English as well as in English – you can Download chapter wise notes, Class 7 Math Notes by Notes Street is prepared by notesstreet Team to help you understand the every part of topic in depth. The notes of Class 7 Hindi And English PDF is your best choice of study material If you are looking out to score high marks. Notes of Class 7 Math Notes In English Chapter – 01 Integers are prepared in accordance with NCERT Syllabus and CBSE guidelines from expert teachers. Notes are as per the NCERT syllabus and guidelines. Each topic in our Chemistry notes are explained as per the exam structure. Along with easy and best explanation we have provided coloured diagrams, images so that topic can be understood in first attempt.. Notes are very useful to score high and colorful diagram is given so it's eye catching and easy to understand. if you want to premium notes Then **Click Here**

Class 7 Math Notes In English Chapter – 01 Integers

Integers

1. Set of positive numbers, 0 and negative numbers is called integers. The set of integers is denoted by I or Z.

2. Natural numbers are contained in the whole numbers.

3. Whole numbers are contained in the integers.

4. Negative numbers are placed on left side of '0' on the horizontal number line. 0 is less than every positive integer and greater than every negative integer.

5. Absolute value of a number is only its numerical value without considering the sign into account.

Properties of integers

i. Closure property is satisfied with respect to addition, subtraction and multiplication of the integers.

For a, b ϵ I, we have:

$a + b \in I$

$a - b \in I$

$a \times b \in I$

ii. Commutative property is satisfied with respect to addition and multiplication of the integers.

If $a, b \in I$, then

$a + b = b + a$

$a \times b = b \times a$

iii. Associative property is satisfied with respect to addition and multiplication of the integers.

If $a, b, c \in I$, then

$a + (b + c) = (a + b) + c = c + (b + a)$

$a \times (b \times c) = (a \times b) \times c = c \times (b \times a)$

iv. Distributive property: Multiplication is distributive over addition and subtraction of the integers.

v. a and -a are additive inverses of each other.

vi. 0 is the additive identity for integers.

vii. 1 is the multiplicative identity for integers.

viii. Division by 0 is not defined.

Adding two positive integers results in positive integers, whereas adding two negative integers will result in the sum with a negative sign. But, the addition of two different signed integers will result in subtraction only and the sign of the result will be the same as the larger number has. See a few examples below:

$2 + 2 = 4$

$2 + (-2) = 0$
$-2 + (-2) = -4$
$-2 - (-2) = 0$

Name	Addition	Subtraction
Closure	$a + b \in z$	$a - b \in z$
Commutative	$a + b = b + a$	$a - b \neq b - a$
Associative	$(a + b) + c = a + (b + c)$	$(a - b) - c \neq a - (b - c)$
Distributive	$a \times (b + c) = ab + ac$	$a \times (b - c) = ab - ac$

Addition and subtraction

Addition and subtraction are two primary arithmetic operations in Maths. Besides these two operations, multiplication and division are also two primary operations that we learn in basic Maths.

The addition represents the values added to the existing value. For example, a basket has two balls, and if we add more than 2 balls to it, there will be four balls in total. Similarly, if there are four balls in a basket and if we take out two balls out of it, then the basket is left with only two balls, which shows subtraction.

Addition and subtraction are not only used for integers but also rational numbers and irrational numbers. Therefore, both the operations are applicable for all real numbers and complex numbers. Also, the addition and subtraction algebraic expressions are done based on the same rules while performing algebraic operations.

Rules to Add and Subtract

Integers are a special group of numbers that are positive, negative and zero, which are not fractions. Rules for addition and subtraction are the same for all.

Negative Sign and Positive Sign

The integers which we add or subtract could be positive or negative.

Hence, it is necessary to know the rules for positive and negative symbols. Positive sign/symbol: (+)

Negative sign/symbol: (-)

Addition of Integers

The three main possibilities in the addition of integers are:

- ➡ Addition between two positive numbers
- ➡ Addition between two negative numbers
- ➡ Addition between a positive number and a negative number

Type of Numbers	Operation	Result	Example
Positive + Positive	Add	Positive (+)	10 + 15 = 25
Negative + Negative	Add	Negative (-)	-10 + (-15) = -25
Positive + Negative*	Subtract	Positive (+)	-10 + 15 = 5
Negative + Positive*	Subtract	Negative (-)	10 + (-15) = -5

Whenever a positive number and a negative number are added, the sign of the greater number will decide the operation and sign of the result. In the above example 10 + (-15) = – 5 and (-10) + 15 = 5; here, without sign 15 is greater than 10 hence, numbers will be subtracted and the answer will give the sign of the greater number.

We know that the multiplication of a negative sign and a positive sign will result in a negative sign, therefore if we write 10 + (-5), it means the '+' sign here is multiplied by '-' inside the bracket. Therefore, the result becomes 10 – 5 = 5.

Alternatively, to find the sum of a positive and a negative integer, take the absolute value ("absolute value" means to remove any negative sign of a number, and make the number positive) of each integer and then subtract these values. Take the above example, 10 + (- 15); absolute value of 10 is 10 and -15 is 15.

⇒ 10 – 15 = -5

Thus, we can conclude the above table as follow:

➥ Adding two positive integers results in positive integer

➥ Adding two negative integers results in a sum of integers with a negative sign.

➥ The addition of a positive and a negative integer gives either a positive or negative sum depending on the value of the given numbers.

Note: The sum of an integer and its opposite is always zero. (For example, -5 + 5= 0)

Subtraction of Integers

Like in addition, the subtraction of integers also has three possibilities. They are:

- ➥ Subtraction between two positive numbers
- ➥ Subtraction between two negative numbers
- ➥ Subtraction between a positive number and a negative number

For ease of calculation, we need to renovate subtraction problems the addition problems. There are two steps to perform this and are given below.

Convert the subtraction sign into an addition sign.

After converting the sign, take the inverse of the number which comes after the sign. Once the transformation is done, follow the rules of addition given above. For example, finding the value of (-5) – (7)

Step 1: Change the subtraction sign into an addition sign

⇒ (-5) + (7)

Step 2: Take the inverse of the number which comes after the sign

⇒ –5 + (-7) (opposite of 7 is -7)

⇒ –5 + (-7) = -12 [Add and put the sign of greater number]

Additive Identity & Additive Inverse

Additive Identity

For every integer a, $a + 0 = 0 + a = a$ here 0 is Additive Identity, since adding 0 to a number leaves it unchanged.

Example: For an integer 2, $2 + 0 = 0 + 2 = 2$.

Additive inverse

For every integer a, $a + (-a) = 0$ Here, $-a$ is additive inverse of a and a is the additive inverse of-a.

Example: For an integer 2, (–2) is additive inverse and for (–2), additive inverse is 2. [Since $+ 2 - 2 = 0$]

Properties of Multiplication of Integers

Closure under Multiplication

For every integer a and b, $a \times b$ = Integer

Commutative Property of Multiplication

For every integer a and b, $a \times b = b \times a$

Multiplication by Zero

For every integer a, $a \times 0 = 0 \times a = 0$

Multiplicative Identity

For every integer a, $a \times 1 = 1 \times a = a$. Here 1 is the multiplicative identity for integers.

Associative property of Multiplication

For every integer a, b and c, $(a \times b) \times c = a \times (b \times c)$

Distributive Property of Integers

Under addition and multiplication, integers show the distributive property. i.e., For every integer a, b and c, $a \times (b + c) = a \times b + a \times c$

These properties make calculations easier.

Multiplication is basically the repeated addition of numbers. For example, if we say, 2 multiplied by 3, it means 2 is added to itself three times.

$2 \times 3 = 2 + 2 + 2 = 6$

Therefore, multiplication of integers is the repeated addition as:

$$a \times n = \underbrace{a + a + a \ldots\ldots a}_{n\ times}$$

Where a and n are both integers.

Division of Integers

When a positive integer is divided by a positive integer, the quotient obtained is a positive integer.

Example: $(+6) \div (+3) = +2$

When a negative integer is divided by a negative integer, the quotient obtained is a positive integer.

Example: $(-6) \div (-3) = +2$

When a positive integer is divided by a negative integer or negative integer is divided by a positive integer, the quotient obtained is a negative integer.

Example: $(-6) \div (+3) = -2$ and

Example: $(+6) \div (-3) = -2$

Introduction to Zero

Integers

Integers are the collection of numbers which is formed by whole numbers and their negatives.

The set of Integers is denoted by Z or I. I = { …, -4, -3, -2, -1, 0, 1, 2, 3, 4,… }

Operations on integers

Addition of integers

i. To add two positive numbers, we add their values and give positive sign to the result. Ex: (+5) + (+2) = +7

ii. To add two negative numbers, we add their values and give negative sign to the result.

Ex: (-12) + (-3) = -15

iii. One positive and one negative number are added by subtracting them and giving the sign of the greater number to result. The greater number is decided by ignoring the signs of the integers.

Ex: (+24) + (-3) = +21

(-32) + (+8) = -24

Subtraction of integers

While subtracting integers, Minuend remains the same and sign of the subtrahend will be changed and then added.

Subtraction is nothing but adding the minuend and subtrahend by changing the sign of the subtrahend.

Ex: (+15) – (+12)

= (+15) + (-12)

(Keep minuend same, change the sign of subtrahend and then add)

= +3

(One +ve and one -ve. Subtract and then give bigger number sign)

Multiplication of integers

To multiply two integers, first signs are multiplied and then the numerals are multiplied. The multiplication of the signs is as follows:

(+) × (+) =+

(-) × (-) =+

(+) × (-) =-

(-) × (+) =-

Division of integers

To divide two integers, first signs will be divided and then the numerals are divided. The division of the signs is as follows:

(+) ÷ (+) = +

(-) ÷ (-) =+

(+) ÷ (-) =-

(-) ÷ (+) =-

6. In an exponential number, base is the number which is repeatedly multiplied. Exponent is the number which tells how many times the base has to be multiplied. In x^7, x is called base and 7 is called exponent or index or power.

7. If the exponent of a negative integer is odd then the value is negative. $(-5)^3 = (-5) \times (-5) \times (-5) = -125$

8. If the exponent of a negative integer is even then the value is positive. 9. Using BODMAS rule, we simplify the numerical expressions.

The order of removing the brackets of an expression is as follows:

() Parathesis, {} Flower or curly brackets and [] Square brackets First we remove () then {} Lastly [].

If a bracket is preceded by a negative term, then on removing bracket the signs of the terms inside the bracket will be changed.

If a bracket is preceded by a positive term, then on removing the bracket the signs of the terms inside the bracket will not change.

Please Share This Share this content

- Opens in a new window

- Opens in a new window

- Opens in a new window

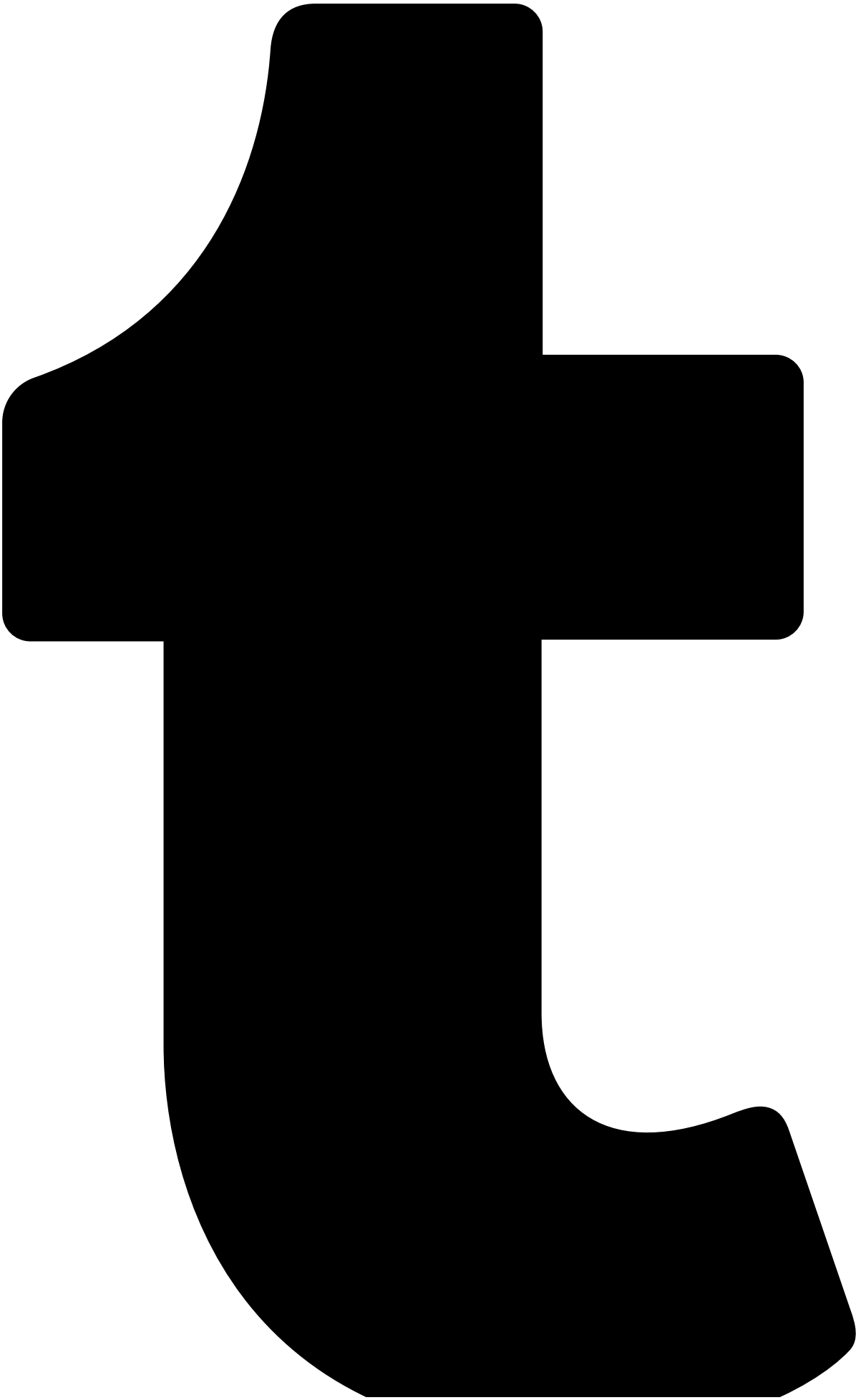
- Opens in a new window

- Opens in a new window

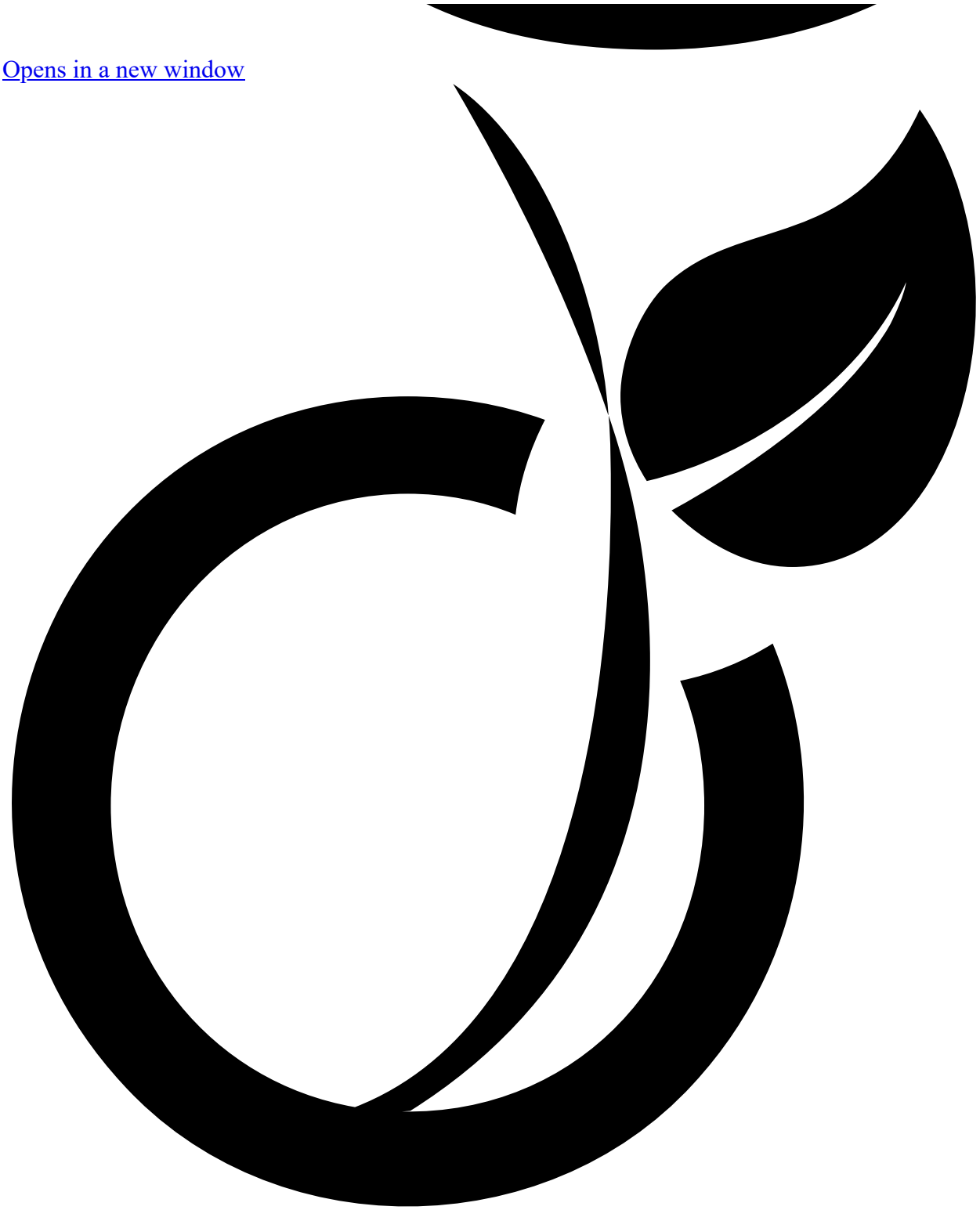
- Opens in a new window

- Opens in a new window

- Opens in a new window

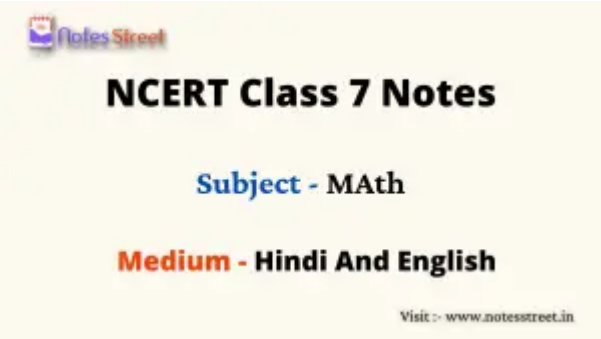


- Opens in a new window



- Opens in a new window

You Might Also Like



NCERT Class 7 Math Notes

June 20, 2022

Class 7 SST Political Science Notes Ch – 09 बाज़ार में एक कमीज़

March 26, 2024

Class 7 भूगोल Notes Ch – 02 हमारी पृथ्वी के अन्दर

March 29, 2024